

Nonlinear-relationship between urban environment and commuting distance considering social and job disparities

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ARTICLE INFO

Keywords:

Commuting distance
Non-linear relationships
Random forest
Housing price
Long-distance commuting

ABSTRACT

Optimizing the built environment to enhance commuting efficiency is promising to alleviate traffic pressure in large cities. However, commuting behavior arises from complex interactions among social and environmental factors, leading to non-linear relationships between commuting distance and its determinants. Using cellphone location-based service data from Shanghai, this study examines the determinants of commuting distance at trip level and the spatial patterns of long-distance commuting at cell level, with particular attention to social disparities. The random forest model combined with SHapley Additive exPlanations (SHAP) analysis are employed to uncover these non-linear effects and interactions. Results show that housing price is the dominating determinant of commuting distance. SHAP analysis further reveals that the interplay between socioeconomic status and urban environment factors drives the observed non-linear impacts on commuting distance. High-skill and high-income people tend to have short commuting trips when they live in central areas or work in outer urban area. Analysis of long-distance commuting patterns indicates that urban structure factors, such as distance to urban center and housing price, exert stronger influence than built environment characteristics. Overall, the findings provide a nuanced understanding of the complex mechanisms underlying commuting behavior and can provide practical implications to sustainable spatial planning.

1. Introduction

The impacts of urban environment on urban commuting behavior has been well-documented in the literature, while their non-linear relationship has not drawn extensive attention until recently (Ewing & Cervero, 2010; Liu et al., 2025). Such complex relationship is the outcomes of spatial and social heterogeneity (Liu & Xiao, 2023; Tong, An, Zhang, et al., 2022). Particularly, scholars are trying to revisit the impacts of social economic status on their commuting for a more nuanced picture (Li et al., 2021; Xiao et al., 2023). The concepts like spatial skills mismatch, describing the phenomenon that people's accessible job opportunities do not fit their working skills, are introduced to the research on urban commuting rather than solely discussing jobs-housing balance (Houston, 2005; Xiao et al., 2023; Zhou et al., 2024). In these studies, how urban environment factors' impacts on urban commuting vary across different population groups have been investigated.

The early spatial mismatch hypothesis is built on the rapid urban

restructuring and argues that low-income people might have longer commuting distance because they cannot move follow jobs due to poor housing affordability (Hu, 2019). As urbanization reaches higher level and inequality keeps rising, commuting behavior becomes more sensitive to people's self-selection on both housing and workplaces based on their socioeconomic status. High housing price would force people away from their workplaces and high-salary would increase labor's preference for long-distance commuting (Dahl & Sorenson, 2010; Xiao, Wei, & Wan, 2021). Also, job accessibility is not the priority in modern urban live and some high-income people prefer outer urban region for better living environment (Qin et al., 2022). Therefore, commuting behavior would be the outcome of the interaction between socioeconomic status and urban structure factors such as housing price and job salary.

Chinese cities usually have strong urban centers with high housing price and high-salary jobs (Wei et al., 2025). People are forced out of the urban center, and only the richest ones can afford living cost there. Such a strong urban center would contribute to many commuting trips

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<https://doi.org/10.1016/j.apgeog.2026.103967>

Received 31 October 2025; Received in revised form 30 January 2026; Accepted 2 March 2026

Available online 4 March 2026

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towards the urban center, which is different from existing knowledge based on western background. Research on commuting behavior in Chinese cities should be tailored to local contexts and socioeconomic status. The nexus between urban structure and socioeconomic status provides a new perspective to explain the complex relationship between urban commuting and its determinants in urban China. Zhou et al. (2024) found that the high-income people in manufacturing usually live in the central urban area and work in the suburban area. Manufacturing jobs are usually suburbanized to avoid negative impacts such as pollution. People in service jobs would live and work in the center urban area, who have short commuting distance. Such job disparities in people's commuting behavior are also highlighted in other studies, which emphasize the importance of professionals and working skills (Kersting et al., 2021; Ling et al., 2024).

As suburbanization and urban expansion intensify, long-distance commuting is becoming increasingly common, and its determinants differ from those of shorter commuting trips. Existing studies have suggested that longer commuting distance is associated with higher mortality risks and causes more stress and anxiety (Sandow et al., 2014; Shitova, 2024). The socioeconomic disparities in commuting may become more pronounced when considering long-distance travel (Huang et al., 2025). Urban environment factors such as built environment and urban structure factors influence people's commuting behavior differently. Built environment factors like pedestrian infrastructure and road network configuration describing micro-level urban landscape and primarily influence short-distance commuting (Ao et al., 2025). By contrast, for long-distance commuting, urban structure factors—such as housing prices, distance to urban centers, and job location—along with socioeconomic status, tend to be more critical determinants (Mitra & Saphores, 2019). Therefore, these differences underscore the importance of paying special attention to long-distance commuting in order to fully understand how built environment, urban structure, and social variables interact to influence commuting behavior.

Insufficient attention has been paid to socioeconomic status or working skills while studying the determinants of commuting behavior. Also, the urban structure factors like housing price are expected to influence long-distance commuting behavior more than the shorter ones (Mitra & Saphores, 2019). However, few studies have clarified the difference in the determinants of short-distance and long-distance commuting. This study examines how urban environment factors—including built environment and urban structure factors—influences urban commuting considering socioeconomic status and job disparities, with particular attention to long-distance commuting. We employ a random forest approach to reveal their non-linear relationships among these determinants. Following research questions are expected to be addressed. (1). How do income and job disparities influence urban commuting? (2). How do the impacts of income and job disparities on commuting distance vary under different urban environment? (3). What are the differences between long-distance commuting and shorter ones in terms of socioeconomic and urban structure factors? These findings contribute to the literature by unfolding the underlying mechanism of the non-linear relationship between urban environment and commuting considering job and social disparities. Moreover, this research illuminates which factors drive long-distance commuting, a critical issue for sustainable urban planning.

2. Literature review

2.1. The determinants of commuting behavior

Numerous studies have demonstrated the impacts of built environment on urban commuting, particularly 3D variables, density, diversity, design. Distance to transit and destination accessibility are incorporated into 3D variables that influence commuting behavior (Ewing & Cervero, 2010). Increasing accessibility to public transportation stations are commonly used to relieve the long-distance commuting problem caused

by jobs-housing imbalance (Blumenberg & King, 2021; Yang & Chang, 2025). Also, destination accessibility influences the spatial layout of residence and job opportunities and indirectly influence urban commuting. Recent literature argues that accessibility to job opportunities is no longer the priority in people's housing choice and people seek for more favorable living or working environment, such as transportation accessibility and entertainment facilities (Li et al., 2019; Wei et al., 2024). Instead, housing affordability is the major barrier for the low and middle income people to decrease commuting distance (Islam & Saphores, 2022; Zhu & Liu, 2025).

Although the impacts of built environment on commuting behavior is a long-lasting topic in urban planning, commuting behavior is still dominated by their socioeconomic status (Chen et al., 2024; Lin et al., 2015). Therefore, spatial skills mismatch has been introduced to the research on commuting, particularly in some cities where urban social inequality keeps rising (Xiao, Wei, & Wan, 2021 & 2023). In general sense, poor people have poor job accessibility because they cannot access proper job opportunities. However, Hu (2019) found that low-income minorities living around the job center in central Los Angeles still have longer commuting distance than the ones in low-density suburbs. Such phenomenon goes beyond just jobs-housing balance or the impacts of built environment, emphasizing social disparities in the research on commuting behavior.

Income and education are the most discussed socioeconomic factors and there are some inconsistent findings. Most studies find that people with lower income usually have longer commuting distance due to their poor housing affordability (Tang et al., 2024). However, urbanization have reversed this preconceived notion that there are studies argued that people commute longer for better paid job (Kersting et al., 2021). People graduated from colleges tend to commute longer than the ones with lower education level (Zhu et al., 2017).

Growing commuting distance calls for more attention on long-distance commuting which have profound impacts on working efficiency and mental health (O'Kelly & Niedzielski, 2009). Most existing studies on long-distance commuting focus on socioeconomic status using survey data. Huang et al. (2025) found that higher income is more associated with long-distance commuting than short-distance commuting. Andersson et al. (2018) found that young people with advanced knowledge-intensive occupations are more like to take long-distance commuting. Therefore, long-distance commuting is more preferred by the commuters have advantages in socioeconomic status because they have better affordability. They take long-distance commuting while they can gain more economic benefit from work (Sandow & Westin, 2014).

Long-distance commuting is generated primarily due to urban structure factors, such as distance to urban center and housing price, which are associated with socioeconomic status. Lower- and middle-income residents are often forced to urban periphery due to the high housing price in the urban center where job opportunities are highly concentrated (Blumenberg & Wander, 2023; Mitra & Saphores, 2019). Meanwhile, urban centers usually have high-salary jobs that attract elite workers who can afford the long-distance commuting (Xiao et al., 2023). Thus, many long-distance commuting trips usually go towards urban centers due to the concentration of high-quality jobs and working environment, but extremely high housing price (Wang et al., 2025). In summary, long-distance commuting is a trade-off between commuting distance and urban living cost, which is highly sensitive to commuter's socioeconomic status and personal preference.

Fig. 1 illustrates a theoretical framework to summarize the determinants of people's commuting behavior, especially long-distance commuting. Jobs-housing relationship is an essential term, which is shaped by built environment. Urban structure factors like distance to urban center, housing price and job salary mediate commuters' housing affordability that further influence long-distance commuting. The impacts of these factors might be modified by commuters' socioeconomic status while commuting behavior is still primarily influenced by

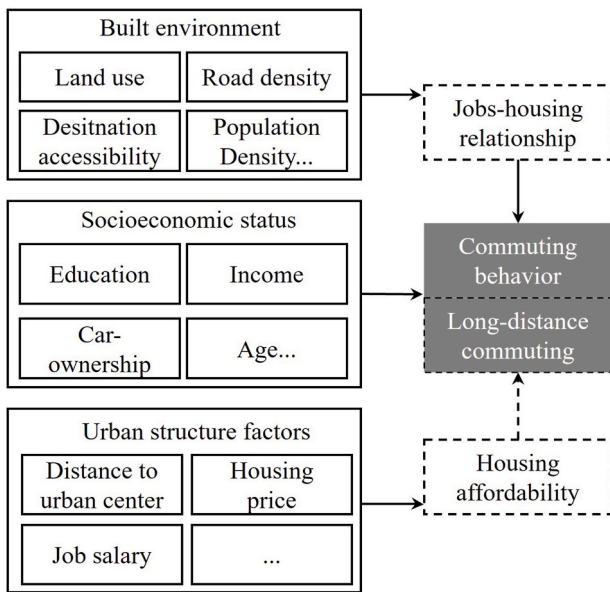


Fig. 1. Theoretical framework for the determinants of commuting behavior.

personal choice. This framework aims to highlight the pathway of how urban structure factors influences long-distance commuting other than jobs-housing relationship.

2.2. Advances in methodologies to address non-linear relationship

Some studies apply machine learning methods to investigate nonlinear features, which can effectively reveal the thresholds of variables and nonlinear patterns (Tong, An, Zhang, et al., 2022). Built environment variables, such as land use mix, transportation configuration, and employment density, have nonlinear impacts on factors like self-containment and commuting burden. In-depth research into these nonlinear features can more accurately present the impacts of built environment, with the goal of optimizing the jobs-housing relationship and improving commuting efficiency.

These non-linear relationships between building environment and commuting trips might be the results of neglecting the impacts of social and job disparities we discuss above. For example, built environment factors have weaker impacts on migrants than local people in the central urban area (Liu & Xiao, 2023). Also, the threshold relationship between public transportation use and daily trip duration is due to the different impacts on low-income people and others (Tao et al., 2023). Therefore, the models that address non-linear relationship should fully consider the interaction between different population groups and built environment.

Explainable machine learning models improve the interpretability and transparency of machine learning models, facilitating the study of the nonlinear relationship between the built environment on jobs-housing relationship. By setting up a SHAP visualization framework, it combines nonlinearity with spatial distribution, providing threshold targets and spatial ranges for refined urban management policy regulation (An, Tong, Tan, et al., 2025). Integrating Random Forest with SHAP algorithms can also reveal the interactions between explanatory variables (Deng, Jiang, Ling, et al., 2025). SHAP partial dependence plots and SHAP interaction values provide insights into the nonlinear relationships and interaction effects between explanatory variables and dependent variables (Ji, Wang, Lyu, et al., 2022).

It is also notable that spatial heterogeneity exists in the research on urban commuting, particularly between core and peripheral areas (Zhao et al., 2024). Although some studies have some initial attempts to explore such relationship, the interaction with social and job disparities has not drawn enough attention (e.g. Liu & Xiao, 2023; Wang et al.,

2025). Existing studies prove that the non-linear relationship exists rather than explaining how it happens. In this research, we would like to provide more explanation about how social and job disparities modify the impacts of urban environment on commuting behavior.

2.3. Summary

Although extensive studies have been implemented on the determinants of commuting behavior, there are still some points worthy of further exploration. First, socioeconomic status would modify the impacts of existing determinants on people's commuting behavior. However, current studies pay less attention to the social disparities in their relationship, making the findings not suitable for some specific population groups (Kersting et al., 2021). Second, the determinants of long-distance commuting trips might be different from the short-distance commuting trips; however, the studies on commuting distance rarely consider the heterogeneity among long-distance and short-distance commuting trips. We argue that urban structure factors influence long-distance commuting trips more than built environment factors. Finally, the traditional statistical tools are not able to provide a comprehensive explanation of the underlying mechanism, while the modern machine learning tools lack explanation from geographical perspective. Therefore, we expect to provide an insight into spatial heterogeneity in the relationship between commuting, urban environment and socioeconomic status via machine learning based models.

3. Study site and data processing

3.1. Study site

Shanghai had over 18% of the commuters with long-distance commuting over 60 min according to the commuting report by Baidu in 2024 (<https://huiyan.baidu.com/boswebsite/cms/report/2025tongqin/>). Social stratification and residential segregation have been well documented in existing literature (Liu et al., 2018; Shen & Xiao, 2020). The socioeconomically disadvantaged people usually commute longer, and there are also significant gaps between different population groups regarding job types, income and education (Yue et al., 2022). Given the long-distance commuting and social segregation, Shanghai is an ideal urban laboratory to study commuting problem and the underlying mechanism (Zhao & Cao, 2020).

In this research, we focus on the main urban area (Fig. 2). Jobs-housing imbalance exists in this region and contributes to long-distance commuting. This region is highly compact and has a mixture of residential, industrial and commercial urban land, including many newly built living communities and development zones. Different kinds of communities are available in this region including high-end commercial center, high-level living communities and urban villages for low-income people. The complex and diverse urban environment contribute to significant job and social disparities, and then long-distance commuting, which can support our research. Long-distance commuting in China usually refers to the trips costing more than 60 min according to the literature (Huang et al., 2025), which corresponds to approximately 15 km in this research.

Here are two scenarios contributing to long-distance commuting trips that might help understand the underlying mechanism. Many low/middle-income workers are forced to live out of the urban center while they still have to work in central urban area, where job opportunities are highly concentrated (Xiao et al., 2023). In this condition, housing price is the primary determinants. On the other hand, advanced job opportunities in Shanghai have experienced suburbanization in recent twenty years (Wei et al., 2025). The talent workers still live in the central urban area for better living environment; however, they have to commute long to outer urban area. The current core-periphery urban structure and recent urban expansion in Shanghai necessarily contribute to many long-distance commuting trips. The high-income and low-income

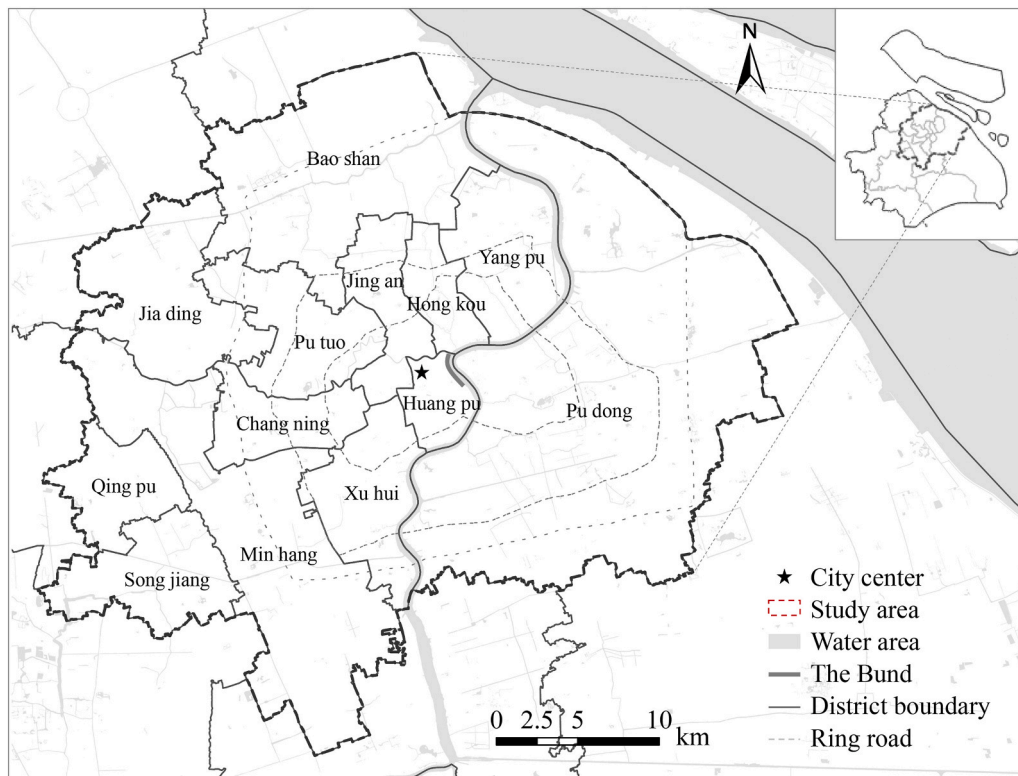


Fig. 2. Study site.

workers have behavior logics when they choose to take long-distance commuting trips.

3.2. Data processing

The location-based service (LBS) data in Shanghai were collected by APP uses from Baidu Inc., covering 80% of smart phone users. The real-time geographic locations of mobile devices with Baidu-use apps were recorded with unique IDs. The LBS data were collected with around 100-m spatial resolution, and were more fine-grained than cellular network data from cellphone companies. This dataset included user profiles such as income and job types which were estimated by their preferences in smart phone use and urban environment around their trajectories (Ling, Niu, Yang, et al., 2024). Leveraging Baidu's machine learning framework, industry sectors for anonymized commuters were identified. This classification integrated large-scale feature tags (such as user portraits, GPS trajectories, and digital behavior) with survey results, point-of-interest (POI) data, and built environment metrics. In this research, we classified the jobs into two main categories: labor-intensive workers included production workers, clerical staff, and service workers; technology-intensive workers included professionals, managers and business owners.

This dataset was suggested to have high accuracy and confirmed as a reliable one by a few studies (Chen et al., 2023; Zhang et al., 2021). The potential residences and workplaces of commuters were inferred from the LBS commuting data based on the spatial-temporal rules (Zhou et al., 2024). The residential location was defined as the primary “anchor point” during non-working hours, and workplace locations was inferred based on daytime stability. The locations of residences and workplaces for total 3,494,277 people and their commuting trips were generated. We tested that the relationship between commuting distance and frequency manifested a log distribution, further validating that our data was convincing.

4. Research methodology

4.1. Variables selection of trip-level and cell-level analyses

In this study, commuting distance was analyzed by explicitly accounting for urban environment features around both residences and workplaces. Based on the conceptual framework (Fig. 1), our regression models incorporated three categories of explanatory variables: built environment, urban structure factors, and socioeconomic status. The variables were chosen on the basis of our insights and the findings of previous research (e.g. Wei et al., 2025; Zhu & Liu, 2025).

Based on existing research on commuting behavior (Ewing & Cervero, 2010), built environment variables were categorized into dimensions of density, diversity, and design. Urban structure variables included housing price and distance to the urban center, which would largely influence living affordability. Socioeconomic status variables were drawn from the LBS dataset and include income and job type. We used job type in place of education level to better capture the skills required in a person's work and, by extension, the opportunities available to them.

Details of dependent and independent variables for the model on commuting distance at trip level are shown in Table 1. We use ratio-based indicators to identify population structure to avoid scale effect because the distribution of population is highly uneven. Also, our data on income is a category data rather than an exact number and high income people refers to the ones whose salary is higher than a threshold. Furthermore, we use difference in housing price between commuting trip origin and destination because we argue that commuting distance is created by the gaps in urban environment, fitting our theoretical framework better (Fig. 1).

Descriptive statistics of the dependent and independent variables used in the trip level analysis of commuting distance are shown in Table 2.

Additional research focused on the ratio of long-distance commuting, which was implemented at 2-km cell level because commuters tended to

Table 1

Details of variables for the model on commuting distance (trip level analysis).

Variables	Data source	Time	Calculation
Dependent variable			
Trip commuting distances	LBS data	2021	Distances of home-to-work trips
Built environment variables			
%Commercial and industrial land around workplace	EULUC-China land use data ^a	2022	Ratio of commercial and industrial land around the destination
%Commercial and industrial land around residence	EULUC-China land use data	2022	Ratio of commercial and industrial land around the origin
Socioeconomic status			
%High-income people	LBS data	2021	Ratio of people with a monthly income exceeding 8000 RMB
%People with technology intensive jobs	LBS data	2021	Ratio of people with technology-intensive job (including professionals, managers, and business owners).
Urban structure variables			
Workplaces' distance to main center (People's Square)	LBS data	2021	Workplaces' distance to main urban center
Residences' distance to main center (People's Square)	LBS data	2021	Residences' distance to main urban center
Difference in housing price	Anjuke (https://www.anjuke.com)	2022	Difference in average housing price between trip origins and destinations

^a Li, Z., & Chen, B. (2024). EULUC China 2.0: Essential Urban Land Use Categories Map across China for 2022 (Shapefile and Geodatabase) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.16794007>.

Table 2

Descriptive statistics of variables for the model on commuting distance at trip level.

Variables	Mean	Standard deviation	Variables	Mean	Standard deviation
Trip commuting distance (km)	5.00	2.56	%People with technology intensive jobs	0.52	0.15
%Commercial and industrial land around workplaces	0.55	0.23	Workplaces' distance to main center (People's Square) (km)	10.33	5.56
%Commercial and industrial land around residences	0.52	0.23	Residences' distance to main center (People's Square) (km)	11.39	5.35
%High-income people	0.35	0.17	Difference in housing price (yuan/m ²)	8834.35	20,693.85

choose nonmotorized travel modes when the commuting distance was within 2-km (Zhou, Yeh, et al., 2022). Also, we compared the difference in the frequency distribution of long-distance commuting between 1-km and 2-km scale (Appendix Figure A2). We found that there were too many cells with only a few long-distance commuting trips, which might influence our analysis results. Therefore, the OD trips were aggregated into 2-km cells based on trip destinations and all the variables in this model were estimated based on the targeted 2-km cells. We aggregated

the data into trip destinations for a work-based commuting trip analysis because we paid particular attention to housing supply in this research. The source and time of the data are available in Table 3, as well as how we estimated these variables.

Descriptive statistics of the dependent and independent variables used in the 2-km cell level analysis of long-distance commuting are shown in Table 4.

4.2. Machine learning based regression model

In this research, a machine learning based regression model is employed to explore the non-linear relationship between commuting behavior and the determinants. The early research on commuting usually use linear model to explore the determinants of commuting behavior. However, threshold-based relationship has drawn much attention in urban research such as the impacts of population density on commuting distance decrease rapidly in high-density region (Liu & Xiao, 2023). Also, we argue that the non-linear relationship is due to the interaction between urban environment and social disparities (Tong et al., 2022).

In order to capture the complex non-linear relationships between commuting behavior and its determinants, this study integrated random forest with SHAP analysis. The hyperparameters of the random forest regression model were carefully tuned to optimize predictive performance. For the number of randomly selected features considered at each split, a value of 4 was selected for both the trip-level and cell-level models, balancing model robustness and computational efficiency. The number of trees was tested from 100 to 1000 in steps of 100, with the optimal values determined as 500 for the trip-level model and 600 for the cell-level model. By incorporating SHAP, the approach not only maintained robust predictive performance but also enabled a transparent decomposition of variable contributions and interaction effects.

The SHAP value evaluates how the presence or absence of a feature affects the prediction considering all possible combinations of features, and fairly distributes the total payoff among players based on their individual contributions. Therefore, it can help explain the contribution of each feature to a particular prediction. The SHAP value for a specific feature is calculated by taking the weighted average of its marginal contributions across all possible feature subsets. Formally, the SHAP value ϕ_i , is:

Table 3

Details of variables for the model on long-distance commuting at 2-km cell level.

Variables	Data source	Time	Calculation
Dependent variables			
%Long-distance commuting trips	LBS data	2021	Ratio of long-distance commuting trips of 2 km cells
Built environment variables			
Floor area ratio	Baidu map	2021	$\frac{\text{Total building area}}{\text{total area}}$
Mixed land use	EULUC-China land use data	2022	$H = - \sum_{i=1}^n P_i \log P_i / \log N$
Distance to subway station	Baidu map	2021	Euclidean distance to nearest subway station
Road density	OpenStreetMap	2021	$\frac{\text{length of roads}}{\text{total area}}$
%Residential land	EULUC-China land use data	2022	$\frac{\text{Area of residential land}}{\text{total area}}$
Urban structure variables			
Distance to main center (People's Square)	LBS data	2021	Euclidean distance to main urban center
Housing price	Anjuke (https://www.anjuke.com)	2022	Average housing price in workplace

Table 4

Descriptive statistics of variables for the model on long-distance commuting at cell level.

Variables	Mean	Standard deviation	Variables	Mean	Standard deviation
%Long-distance commuting trips	0.15	0.05	%Residential land	0.41	0.25
Floor area ratio	1.16	0.50	Distance to main center (People's Square) (km)	14.32	5.30
Mixed land use	0.71	0.10	Housing price (yuan/m ²)	53,175.21	21,260.65
Distance to subway station (km)	1.46	1.06	%High-income people	0.31	0.08
Road density (km/km ²)	1.10	0.48	%People with technology intensive jobs	0.48	0.05

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (1)$$

where N is the total number of input features. S is a subset of features that does not include feature i . $f(S)$ is the model output when only the features in subset S are used. $f(S \cup \{i\})$ is the model output when the features in subset S and feature i are simultaneously used. $|S|$ is the number of features in subset S . $|N|$ is the total number of features.

So far, the models for addressing non-linear relationship have gained currency and been proved to be powerful in research on commuting behavior. In this research, we construct an interpretable nonlinear machine learning based method. Partial dependence plot (PDP) is employed to identify the non-linear relationship between explanatory variables and urban commuting. Furthermore, we introduced an interaction term into the model focusing on the interaction between socio-economic status and urban environment. The results are expected to unravel how the determinants of commuting distance vary across

different population groups.

5. Urban commuting in Shanghai: social disparities and spatial heterogeneity

5.1. Spatial pattern of urban commuting at trip level

The spatial distribution of people's commuting trips is visualized using community detection method. Community detection can help identify the regions that well connected by commuting trips. It is helpful to detect self-contained regions and provide an overview of the connection between regions from commuting OD network. It has been widely used in research on urban mobility (Xiao et al., 2024; Zhang et al., 2023), and the details of the commuting detection method could be found in reference (Wang, Zhou, Rong, et al., 2022).

Fig. 3 shows the results of commuting detection model and their linkage via commuting flows. The arrows present the volume of commuting flows and the circles represent the number of residents. Unlike the highly concentrated urban centers of housing price or job opportunities in Shanghai (Li et al., 2019; Li et al., 2024; Wei et al., 2024), commuting network is highly polycentric. There are 16 communities based on people's commuting trips which can be deemed as self-contained regions. There were strong linkages connected between these communities with their neighbors, that would not contribute to many long-distance commuting trips.

5.2. Spatial distribution of key variables at the 2-km cell level

In this part, we mapped some key internal and external variables for modeling analysis in Fig. 4. Fig. 4 (a) and (b) separately presents the spatial distribution in the rate of residential space and housing price. The urban center does not have highest rate of residential space, but have highest housing price in Shanghai, including the new downtown in Pudong district. The residential center in Shanghai is located in the periphery of the central urban area, where housing price is more affordable and the distance to the main urban center is proper. The young and talent people prefer to live in the periphery of urban center where living environment is better and the resources for work and

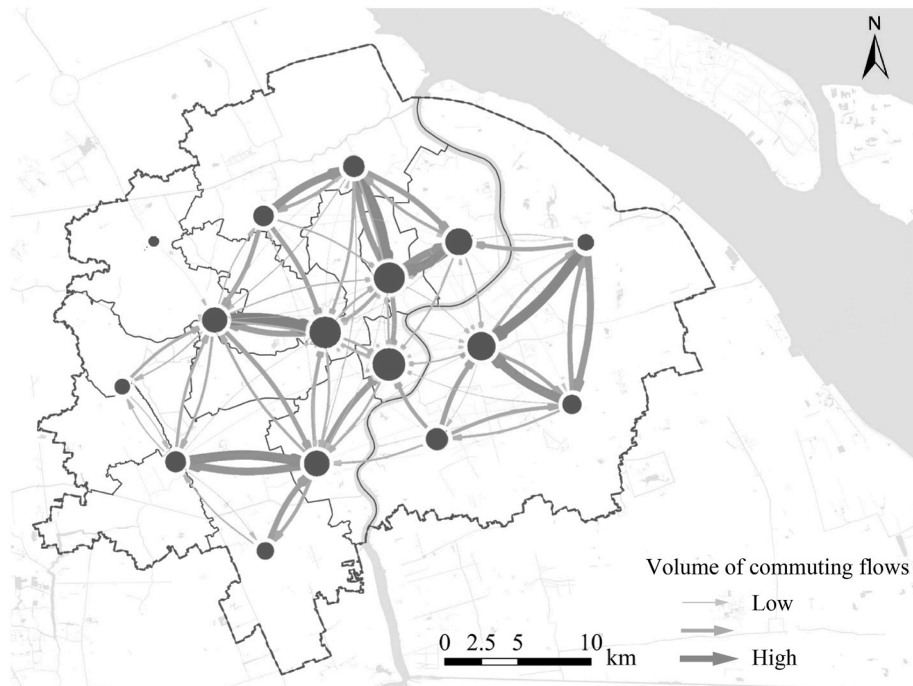


Fig. 3. Commuting pattern based on community detection method.

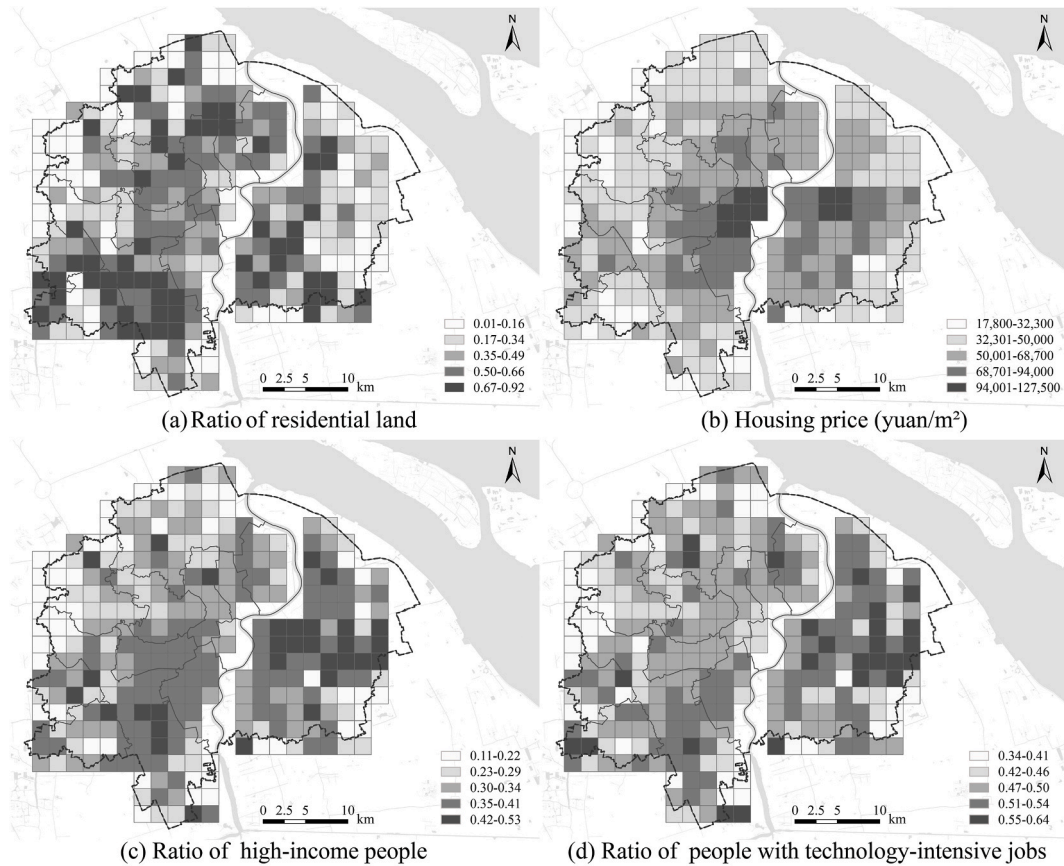


Fig. 4. Key variables in this research.

leisure are ample (Shen & Xiao, 2020; Xiao et al., 2023). The main urban center is still for the advanced producer service like financial service that would significantly raise rental cost (Li et al., 2024; Wei et al., 2024).

Fig. 4(c) and (d) separately presents the distribution of people with high-income and technology-intensive jobs. The western part of Shanghai is the traditional central urban area and there is a concentration of high-income people because many producer service industries are still there (Wei et al., 2025). In the eastern part of Shanghai, there are special economic zones with high-tech parks, attracting many technology-intensive jobs (Reardon, 1998). Such an open-door policy helps the eastern part of Shanghai attract more high-income people than the traditional central urban area.

6. Results of regression models

6.1. Spatial and social disparities in determinants of commuting at trip level

The determinants of commuting behavior were analyzed at trip level, considering the urban environment around both origins and destinations of trips. We compared the performance of different machine learning based models. The random forest model achieved the high predictive accuracy (0.68) and Gradient Boosting Decision Tree (GBDT, 0.67) demonstrated comparable performance. In addition, we found that the output of GBDT in this model was unstable and random forest was more robust in noise resistance, overfitting prevention, and processing non-standardized data. Therefore, the random forest model was adopted as a robust model for further analysis.

Table 5 indicates that housing price is the most important determinant of commuting distance in Shanghai, with a relative importance of 33.39%. Housing price serves as a comprehensive proxy for the quality of the urban living environment, reflecting multiple dimensions such as

Table 5

Importance of the variables for the model at trip level.

Variables	Relative importance	Rank
Difference in housing price	33.39%	1
Workplaces' distance to main center	21.24%	2
Residences' distance to main center	17.63%	3
%People with technology intensive jobs	8.19%	4
%Commercial and industrial land around workplace	6.80%	5
%High-income people	6.57%	6
%Commercial and industrial land around residence	6.18%	7

public facilities, street design, and other neighborhood attributes. Differences in housing prices therefore imply disparities in the residential environment between trip origins and destinations.

Our results further suggest that when the housing price gap is below 25,000 yuan/m², housing price is not strong enough to induce longer commuting trips, which are instead shaped by other factors (Fig. 5a). When housing prices at trip origins are more than 25,000 yuan/m² lower than those at destinations, individuals tend to commute longer because they cannot afford the high housing costs near their workplaces (Xiao et al., 2023). Conversely, when housing prices at trip origins are more than 25,000 yuan/m² higher than those at destinations, it is likely that the urban design quality, access to public facilities around workplaces do not meet their needs (Li et al., 2019). In such cases, people may accept longer commutes in exchange for a better living environment.

Fig. 5 (b) provide insight into the disparities in people's commuting distance regarding income. Generally, high-income people have higher housing affordability that have better access to job opportunities. However, the plots in Fig. 5 manifest non-linear relationship between income and urban commuting. According to Fig. 5 (b), residents' commuting distance increased as their income raise and increasing rate

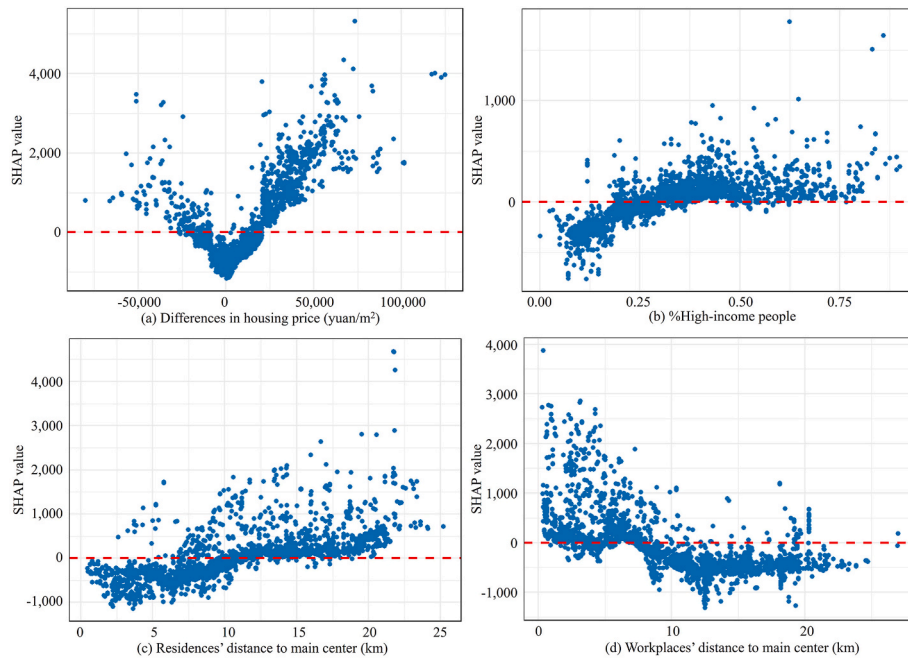


Fig. 5. Results of the random forest model.

is highest when their income is at medium level. Therefore, housing affordability might not be able to explain people's commuting distance because income is positively associated with commuting distance. We detect a threshold-based relationship that people's commuting distance would increase sharply when the proportion of high-income people reach 0.3. Although this threshold does not have practical meaning, it still could be concluded that, people can endure long distance commuting when income reach a threshold.

Distance to the urban center is the other well-discussed variable in commuting research, and residences' and workplaces' distance to urban centers are important variables just next to differences in housing price. Fig. 5 (c) and (d) manifest a non-linear relationship between commuting distance and residences' and workplaces' distance to urban center, respectively. When the distance to urban center was lower than 10 km, residences' distance to urban center would decrease commuting distance. A similar threshold effect was observed for workplaces, where

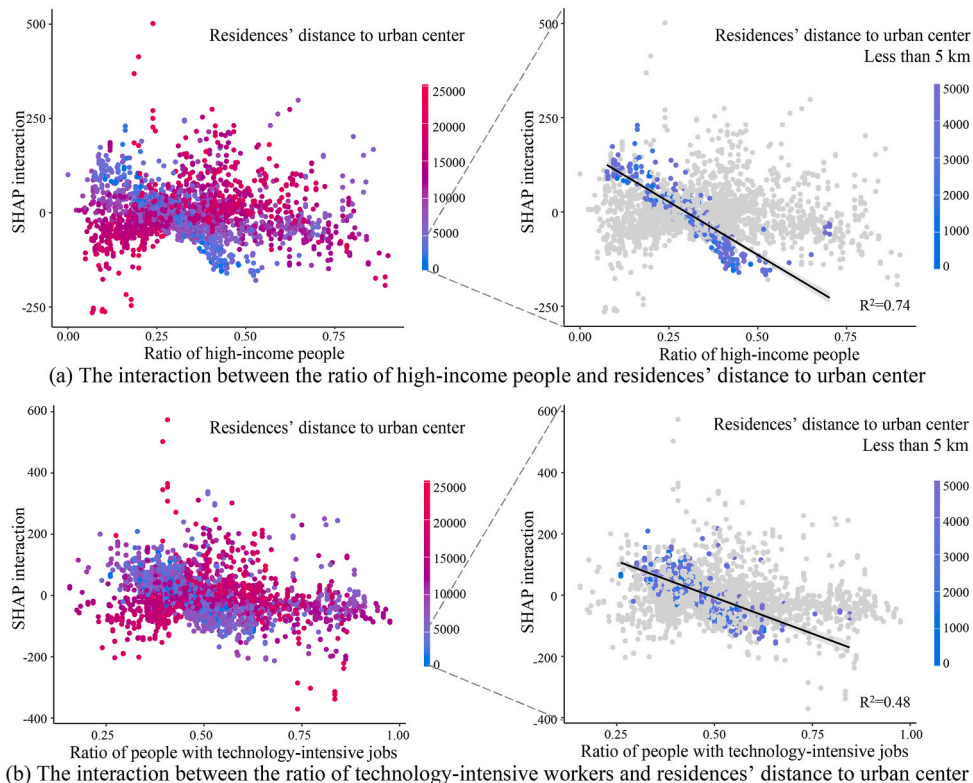


Fig. 6. The interaction between socioeconomic factors and distance to urban center based on partial dependence plot.

commuting distance increased when the workplace was within 8 km of the urban center and decreased beyond this range. However, when the distance to urban center was higher than 8 km, workplaces' distance to urban center would decrease commuting distance. These non-linear relationships suggested that there were thresholds in the relationship between proximity to urban center and commuting distance in Shanghai.

We further explored the impacts of the interaction between socio-economic factors and distance to urban center on people's commuting behavior (Fig. 6). The y-axis represents commuting distance and x-axis represents the variables. The color of the points represents the distance to urban center. The left figures present the results for all the regions and the right ones present results for the cells within a certain distance to the urban center. Fig. 6 (a) shows the interaction between the ratio of high-income people and residences' distance to urban center. It is intriguing that there is a significant linear relationship between high-income people's commuting distance and their residence to urban center when the distance to urban center is smaller than 5 km. The high-income people have significant advantages in residence choice in the urban center and proximity to workplaces is a critical concern. Fig. 6 (b) shows that when the workplaces are more than 15 km away from the urban center, higher ratio of high-income people can reduce commuting distance. Since commuting behavior is the balance between proximity to workplace and living environment, high-income people might not be able to find living environment good enough to compensate for the long commuting distance. They might live close to their workplace while they have high housing affordability and flexible in housing choice. Synthesizing these analyses, proximity to urban center is not the priority for high-income and people's residential choice in the regions 15 km away from the urban center.

6.2. Social and spatial disparities in determinants of long-distance commuting trips at the cell level

An additional work-based commuting analysis is applied on the percentage of long-distance commuting to identify the determinants. Our analytical framework argues that long-distance commuting is primarily determined by urban structure attributes and personal socioeconomic status rather than built environment (Fig. 1). The analytical results support our argument that housing factors and socioeconomic factors are more important than the built environment factors like road network and mixed land use (Table 6).

The ratio of long-distance commuting is closely associated with the ratio of residential land near workplaces, referring to housing supply. As shown in Fig. 7 (a), a nonlinear relationship is observed between the ratio of residential land and the long-distance commuting trips. Specifically, when the ratio of residential land is below 0.5, it contributes to a higher proportion of long-distance commuting. Conversely, an increase in residential land supply is associated with a reduction in long-distance commuting. Therefore, increasing housing supply can significantly reduce long-distance commuting, highlight the importance of jobs-housing balance.

The importance of housing price has decrease to 12.74% compared

Table 6
Importance of the variables for the model on long-distance commuting.

Variables	Relative importance	Rank
%Residential land	15.50%	1
%High-income people	14.00%	2
Housing price	12.74%	3
%People with technology-intensive jobs	12.50%	4
Road density	12.38%	5
Distance to subway station	9.22%	6
Mixed land use	8.55%	7
Distance to main center	7.68%	8
Floor area ratio	7.41%	9

to its relative importance in Table 5. According to Fig. 7 (b), higher housing price significantly contributes to higher ratio of long-distance commuting trips when housing price exceed 50,000 yuan/m². This finding corresponds to findings in literature that housing affordability is a critical trigger of long-distance commuting (Blumenberg & Wander, 2023; Mitra et al., 2019). Generally, higher housing price is associated with better accessibility to job opportunities, and it explain why high housing price decrease ratio of long-distance commuting when housing prices are lower than 50,000 yuan/m².

Housing affordability would be modified by commuters' socioeconomic status, such as high-income people would be more flexible in housing selection (Xiao et al., 2023). However, Fig. 7 (c) suggests that the regions with higher income would like to have more long-distance commuting trips. Housing price in Shanghai is still a challenge to workers that people have to commute long for higher income. On the other hand, the high-skill people manifest distinguished patterns that higher ratio of high-skill people have lower ratio of long-distance commuting trips (Fig. 7 (d)). High-skill jobs are usually highly concentrated and long-distance commuting might influence high-skill workers' efficiency (Parenti & Tealdi, 2019). This is a new insight into the job disparities in people's commuting behavior, rather than only consider socioeconomic status.

7. Discussion

Rising commuting distance has become an urban challenge and much attention has been paid to how to optimize urban land use and structure to reduce commuting distance. In this research, people's commuting behavior is analyzed in both trip and cell levels to explore the impacts of urban environment and the heterogeneity in socioeconomic status. Particular attention has been paid to the interaction between urban environment and socioeconomic status because we emphasize the perspective of commuters' self-selection. There are several findings worthy of further discussion.

Regarding commuting distance, this research considers built environment around both origins and destinations of commuting trips, primarily focusing on job and housing supply, and urban structure factors including housing price and distance to urban center. One of our key findings is that housing price emerges as the most critical determinant of commuting distance in Shanghai. Spatial mismatch hypothesis has dominated research on commuting for many years, which emphasizes jobs-housing balance in influencing commuting behavior (Blumenberg & King, 2021). In the large metropolis like Shanghai, housing affordability has become a pressing issue: many households cannot afford housing in central or well-served areas (Li et al., 2019). Blumenberg and Wander (2023) explained such phenomenon via the mismatch between low-wage jobs and high rental or house costs, which are suggested as the fundamental trigger of longer commuting distance in high-cost metropolis. Based on these insights, we argue that urban structure factors, especially housing cost and job salary, deserve greater attention in commuting research.

Residences' and workplaces' distance to urban center are also critical determinants; however, we identify non-linear relationship between distance to urban center and commuting distance. Distance to urban center only contribute to commuting distance when the value is more than 6 km in this research. Such non-linear relationship is an addition to existing literature relaying on core-periphery pattern when explaining commuting behavior (Copus, 2001; Xiao et al., 2023; Zhao et al., 2024). Therefore, urban center should not be treated as a point, but as a space, because the non-linear relationship suggests that there is significant heterogeneity between central urban region and others.

The findings regarding the interaction between socioeconomic status and distance to urban center provide a deeper understanding of the heterogeneity in central urban area. Taking 5 km to urban center to define central urban area, we found that high-income and high-skill people have short commuting distance when they live in central urban

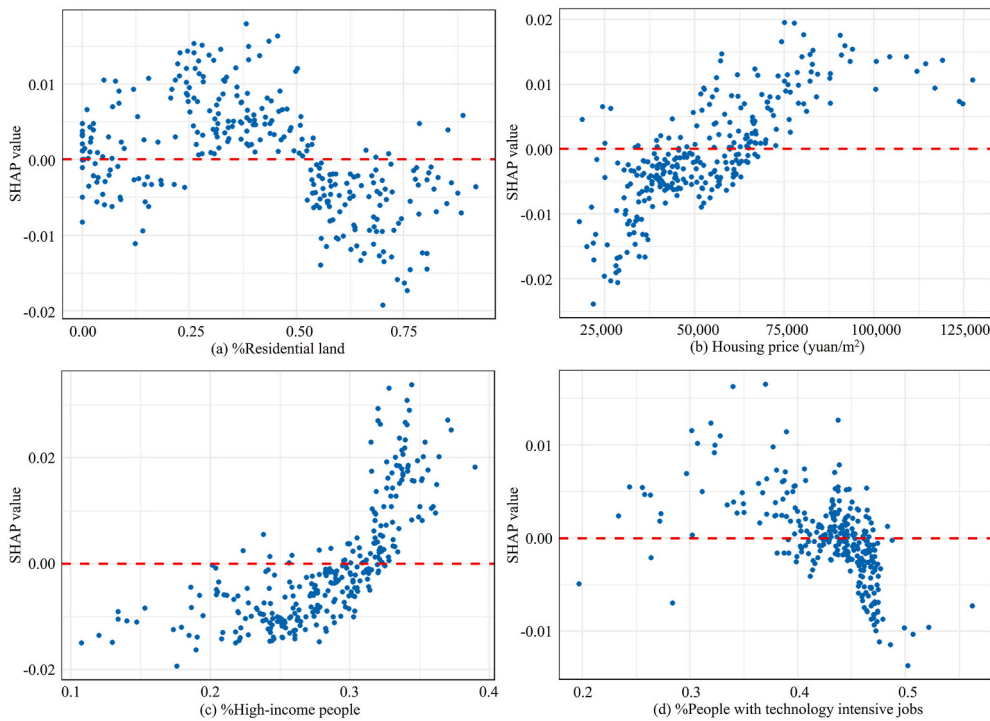


Fig. 7. Partial dependence plot of the variables in the model of long-distance commuting.

area or work in outer urban area. These findings correspond to some results that the low-income residents in central urban area commute longer (Lin et al., 2015; Shen, 2000). When high-skill and high-income people can afford housing price in urban center, they can access suitable jobs easily because central urban is a concentration of high-salary and high-skill jobs (Wei et al., 2024). When high-salary and high-skill people work in the central urban area, some of them might choose to live outside for better living environment with proximity to green space (Qin et al., 2022). High-skill and high-income people are flexible in housing choice and some of them move to urban periphery following the suburbanized job opportunities (Wei et al., 2024; Zhu & Wang, 2022). The interaction between socioeconomic status and distance to urban center further highlight heterogeneity in the impacts of housing affordability and skill mismatch in shaping commuting distance between core region and outer urban region (Mitra & Saphores, 2019; Xiao et al., 2023).

This research pays particular attention to the determinants of long-distance commuting, which are distinguished from regular commuting. Our results suggest that, housing affordability associated factors such as housing supply, housing price and people socioeconomic status are more important determinants of the ratio of long-distance commuting than urban environment factors like mixed land use and road density. Since built environment factors have been more emphasized in current research on commuting (Liu & Xiao, 2023; Tong et al., 2022), we call for more efforts on urban structural factors such as housing price.

On the other hand, we found that when the proportion of residential land is more than 50%, long-distance commuting would be significantly reduced. Also, higher housing price would contribute to long-distance commuting when housing price is higher than 50,000 yuan/m², lower than average housing price in Shanghai. Therefore, housing availability and affordability are both important to explain long-distance commuting while commuting distance is more sensitive to housing price. This finding provide an insight into the determinants of long-distance commuting and different underlying mechanism from commuting distance, which are worthy of further exploration (Wang et al., 2025; Xiao et al., 2023).

These findings also provide implication to urban practice. First,

urban planning in Shanghai should pay more attention to housing affordability rather than solely focusing on jobs-housing balance. Affordable housing is a strategy to reduce low/middle-income workers' commuting distance in outer urban area. Also, high-quality facilities including medical service, education and entertainment should be decentralized that help high-skill workers suburbanize following their jobs. Our PDP analysis suggests that low-income and low-skill workers living within 5 km and work out of 15 km to the urban center is worthy of particular attention. The heterogeneity in commuting behavior across different social economic groups require planning policies better fit local context.

Finally, we would like to thank the advanced machine learning models that can further explore non-linear relationships and interaction between variables that earlier studies were unable to detect (Ji et al., 2022; Liu & Xiao, 2023; Tao, 2023; Tong et al., 2022). In this research, there is a complex interaction between urban environment, socioeconomic status and commuting behavior. The machine learning tools help unravel the heterogeneity regarding location and socioeconomic status when studying the relationship between urban environment and commuting behavior. Also, the LBS data on commuting behavior and the technology to estimate socioeconomic status are important extensions of traditional travel survey data, as a promising data source in the research on commuting. Leveraging these tools, our study presents a more nuanced picture that how urban environment, socioeconomic status, and job factors interact in shaping commuting distances.

Limitations also exist in this research. First, the commuting flow in this research does not include socioeconomic status. The error might be more evident when the travel volume is high and population groups are highly mixed. This is a limitation of current phone signal data because the data provider in China can only provide aggregated data to protect private policy. Also, our study area does not cover the whole Shanghai, making the core-periphery heterogeneity not evident as other studies. Even though, we still identify the phenomenon that high-income people with technology intensive jobs would have shorter commuting distance. Furthermore, this research only identifies the interaction between distance to urban center and socioeconomic status, while other built environment and urban structure factors are worthy of further

investigation.

8. Conclusions

This study investigates the non-linear determinants of commuting distance in Shanghai using cellphone-based commuting trajectories, integrating socioeconomic attributes, built environment characteristics, and urban spatial structure factors. The results demonstrate that housing affordability is the most influential driver of commuting distance, indicating that urban structure constraints can dominate commuting outcomes in high-cost metropolitan contexts. Particularly, urban structure factors are proved to be more important determinants of long-distance commuting trip generation than built environment factors such as mixed land use and street design.

Further analysis reveals the complex interaction between socioeconomic status and distance to urban center when they influence commuting distance, corresponding to their non-linear relationship. We find that high-income and high-skill people would have shorter commuting trips when they live in central urban area or work in the outer urban area. In this research, 5 km to the urban center can be employed to define the central urban area, where heterogeneity is identified. These findings manifest the fact that the well-discussed relationship between urban center accessibility and commuting distance has been modified by socioeconomic status. As urban and social inequality keep rising, self-selection based on commuters' socioeconomic status should be emphasized in future research on urban commuting.

CRedit authorship contribution statement

Xingang Zhou: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Zhouye Zhao:** Visualization, Software, Methodology, Formal analysis, Data curation. **Weiye Xiao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Weiye Xiao reports financial support was provided by National Natural Science Foundation of China. Xingang Zhou reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We would like to acknowledge the funding of National Natural Science Foundation of China, Grant No. 42571278, 42201231, the Fundamental Research Funds for the Central Universities (2025-1-ZD-04).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apgeog.2026.103967>.

References

An, R., Tong, Z., Tan, B., Xiong, Q., Luo, Y., Liu, Y., ... Yang, X. (2025). Revealing the relationship between 2D/3D built environment and jobs-housing separation coupling nonlinearity and spatial nonstationarity. *Journal of Transport Geography*, 123, Article 104112.

- Andersson, M., Lavesson, N., & Niedomysl, T. (2018). Rural to urban long-distance commuting in Sweden: Trends, characteristics and pathways. *Journal of Rural Studies*, 59, 67–77.
- Ao, Y., Long, Y., Zheng, J., & Bahmani, H. (2025). The influence of rural built environment on residents' non-commuting travel mode choices. *Journal of Transport Geography*, 128, Article 104330.
- Blumenberg, E., & King, H. (2021). Jobs-housing balance re-visited. *Journal of the American Planning Association*, 87(4), 484–496.
- Blumenberg, E., & Wander, M. (2023). Housing affordability and commute distance. *Urban Geography*, 44(7), 1454–1473.
- Chen, L., Chen, M., & Fan, C. (2024). Age disparities and socioeconomic factors for commuting distance in Beijing by explainable machine learning. *Cities*, 155, Article 105493.
- Chen, R., Zhang, M., & Zhou, J. (2023). Jobs-housing relationships before and amid COVID-19: An excess-commuting approach. *Journal of Transport Geography*, 106, Article 103507.
- Copus, A. K. (2001). From core-periphery to polycentric development: Concepts of spatial and aspatial peripherality. *European Planning Studies*, 9(4), 539–552.
- Dahl, M. S., & Sorenson, O. (2010). The social attachment to place. *Social Forces*, 89(2), 633–658.
- Deng, Y., Jiang, W., Ling, Z., Liu, L., & Sun, S. (2025). Revealing the driving factors of urban wetland park cooling effects using Random Forest regression and SHAP algorithm. *Sustainable Cities and Society*, 120, Article 106151.
- Ewing, R., & Cervero, R. (2010). Travel and the built environment: A meta-analysis. *Journal of the American Planning Association*, 76(3), 265–294.
- Houston, D. (2005). Employability, skills mismatch and spatial mismatch in metropolitan labour markets. *Urban Studies*, 42(2), 221–243.
- Hu, L. (2019). Racial/ethnic differences in job accessibility effects: Explaining employment and commutes in the Los Angeles region. *Transportation Research. Part D, Transport and Environment*, 76, 56–71.
- Huang, J., Fan, W., Wang, J., Loo, B., & Qi, W. (2025). Long commuting, social inequality, and urban development with scaling laws in China. *The Innovation Geoscience*, 3(3), 100164-1.
- Islam, M. R., & Saphores, J. D. M. (2022). An LA story: The impact of housing costs on commuting. *Journal of Transport Geography*, 98, Article 103266.
- Ji, S., Wang, X., Lyu, T., Liu, X., Wang, Y., Heinen, E., & Sun, Z. (2022). Understanding cycling distance according to the prediction of the XGBoost and the interpretation of SHAP: A non-linear and interaction effect analysis. *Journal of Transport Geography*, 103, Article 103414.
- Kersting, M., Matthies, E., Lahner, J., & Schlüter, J. (2021). A socioeconomic analysis of commuting professionals. *Transportation*, 48, 2127–2158.
- Li, H. (2024). Geography of housing bubbles in Shanghai: A study of price-to-rent ratios at the community level. *International Journal of Housing Policy*, 1–24.
- Li, T., Dodson, J., & Goldie, X. (2021). Urban structure, commuting burden, and employment status of labour forces in an Australian city. *Journal of Transport Geography*, 93, Article 103072.
- Li, H., Wei, Y. D., & Wu, Y. (2019). Urban amenity, human capital and employment distribution in Shanghai. *Habitat International*, 91, Article 102025.
- Lin, D., Allan, A., & Cui, J. (2015). The impacts of urban spatial structure and socioeconomic factors on patterns of commuting: A review. *International Journal of Urban Sciences*, 19(2), 238–255.
- Ling, C., Niu, X., Yang, J., Zhou, J., & Yang, T. (2024). Unravelling heterogeneity and dynamics of commuting efficiency: Industry-level insights into evolving efficiency gaps based on a disaggregated excess-commuting framework. *Journal of Transport Geography*, 115, Article 103820.
- Liu, L., Huang, Y., & Zhang, W. (2018). Residential segregation and perceptions of social integration in Shanghai, China. *Urban Studies*, 55(7), 1484–1503.
- Liu, Y., Shen, R., He, M., Li, X., & Shi, Z. (2025). Gender differences in commuting travel mode choices among young adults: A spatial heterogeneity perspective. *Journal of Transport Geography*, 123, Article 104145.
- Liu, J., & Xiao, L. (2023). Non-linear relationships between built environment and commuting duration of migrants and locals. *Journal of Transport Geography*, 106, Article 103517.
- Mitra, S. K., & Saphores, J. D. M. (2019). Why do they live so far from work? Determinants of long-distance commuting in California. *Journal of Transport Geography*, 80, Article 102489.
- O'Kelly, M. E., & Niedzielski, M. A. (2009). Are long commute distances inefficient and disorderly? *Environment & Planning A*, 41(11), 2741–2759.
- Parenti, A., & Tealdi, C. (2019). The role of job uncertainty in inter-regional commuting: The case of Italy. *Growth and Change*, 50(2), 634–671.
- Qin, X., Wei, Y. D., Yu, Z., & Xiong, N. (2022). Urbanization, suburbanization, and population redistribution in urban China: A case study of Nanjing. *Journal of Urban Planning and Development*, 148(4), Article 05022034.
- Reardon, L. C. (1998). Learning how to open the door: A reassessment of China's "Opening" strategy. *The China Quarterly*, 155, 479–511.
- Sandow, E., Westerlund, O., & Lindgren, U. (2014). Is your commute killing you? On the mortality risks of long-distance commuting. *Environment & Planning A*, 46(6), 1496–1516.
- Shen, Q. (2000). Spatial and social dimensions of commuting. *Journal of the American Planning Association*, 66(1), 68–82.
- Shen, J., & Xiao, Y. (2020). Emerging divided cities in China: Socioeconomic segregation in Shanghai, 2000–2010. *Urban Studies*, 57(6), 1338–1356.
- Shitova, Y. Y. (2024). The impact of long-distance travel to work on the health of commuting labour migrants: A literature review. *Population and Economics*, 8(1), 37–51.

- Tang, J., Zhang, S., & Zhang, M. (2024). The impact of perceived income level on commuting time: Evidence from Shanghai, China. *Travel Behaviour and Society*, 37, Article 100862.
- Tao, S., et al. (2023). Examining the non-linear effects of transit accessibility on daily trip duration: A focus on the low-income population. *Journal of Transport Geography*, 109, Article 103600.
- Tong, Z., An, R., Zhang, Z., Liu, Y., & Luo, M. (2022). Exploring non-linear and spatially non-stationary relationships between commuting burden and built environment correlates. *Journal of Transport Geography*, 104, Article 103413.
- Wang, J., Sun, Z., Cao, J., Liu, K., & Wang, D. (2025). Counterproductive development: New towns may produce and attract more long-distance commuting. *Transportation Research Part D: Transport and Environment*, 147, Article 104958.
- Wei, Y. D., Xiao, W., & Wu, Y. (2024). Centring or suburbanization? Changing locations of producer services in Shanghai. *Environment and Planning A: Economy and Space*, 56 (5), 1482–1502.
- Wei, Y. D., Xiao, W., & Wu, Y. (2025). Polycentric urban development in China: Dimensions, effects, and policies. *Chinese Geographical Science*, 1–15.
- Xiao, W., Li, H., Yuan, F., & Chen, W. (2024). The 2-month lockdown and commuting behavior transformation in Shanghai: A phone signal data-based analysis. *Transport Policy*, 150, 149–161.
- Xiao, W., Wei, Y. D., & Chen, W. (2023). Skills mismatch, jobs-housing relationship and urban commuting. *Travel Behaviour and Society*, 33, Article 100610.
- Xiao, W., Wei, Y. D., & Li, H. (2021a). Spatial inequality of job accessibility in Shanghai: A geographical skills mismatch perspective. *Habitat International*, 115, Article 102401.
- Xiao, W., Wei, Y. D., & Wan, N. (2021b). Modeling job accessibility using online map data: An extended two-step floating catchment area method with multiple travel modes. *Journal of Transport Geography*, 93, Article 103065.
- Yang, W., & Chang, J. S. (2025). A quasi-experimental study of light rail transit on jobs-housing balance by regional typology: A case study of South Korea. *Journal of Transport Geography*, 124, Article 104173.
- Yue, L., O'Kelly, M. E., & Wu, R. (2022). An empirical study of commuting efficiency between different educational categories of workers in Shanghai. *Geographical Analysis*, 54(4), 820–838.
- Zhang, J., Tan, S., Peng, C., Xu, X., Wang, M., Lu, W., ... Yu, H. (2023). Heterogeneous changes in mobility in response to the SARS-CoV-2 Omicron BA. 2 outbreak in Shanghai. *Proceedings of the National Academy of Sciences*, 120(42), Article e2306710120.
- Zhang, Y., Zhang, Y., & Zhou, J. (2021). A novel excess commuting framework: Considering commuting efficiency and equity simultaneously. *Environment and Planning B: Urban Analytics and City Science*, 48(1), 151–168.
- Zhao, P., & Cao, Y. (2020). Commuting inequity and its determinants in Shanghai: New findings from big-data analytics. *Transport Policy*, 92, 20–37.
- Zhao, P., Wang, H., Liu, Q., et al. (2024). Unravelling the spatial directionality of urban mobility. *Nature Communications*, 15, 4507.
- Zhou, X., Sun, C., Niu, X., & Shi, C. (2022a). The modifiable areal unit problem in the relationship between jobs–housing balance and commuting distance through big and traditional data. *Travel Behaviour and Society*, 26, 270–278.
- Zhou, L., Xiao, W., Li, H., Wang, C., Wang, X., & Zheng, Z. (2024). Identify social and job disparities in the relationship between job-housing balance and urban commuting using Baidu trajectory big data. *Transactions in GIS*, 28(2), 258–277.
- Zhou, X., Yeh, A. G. O., Yue, Y., & Li, W. (2022b). Residential-employment mixed use and jobs-housing balance: A case study of Shenzhen, China. *Land Use Policy*, 119, Article 106201.
- Zhu, Z., Li, Z., Liu, Y., Chen, H., & Zeng, J. (2017). The impact of urban characteristics and residents' income on commuting in China. *Transportation Research. Part D, Transport and Environment*, 57, 474–483.
- Zhu, Y., & Liu, S. (2025). Linking housing prices and innovation: The role of commuting distance. *Applied Spatial Analysis and Policy*, 18(1), 26.
- Zhu, Y., & Wang, D. (2022). Growing threat to urban retail? Residential suburbanization and shopping behavior change in Shanghai, China. *Cities*, 131, Article 104029.